Subject: Manufacturing Processes (23A0340IT)

UNIT - II : WELDING

Classification, welded joints, gas welding & cutting, arc welding (MMAW, SAW, TIG, MIG, ESW), resistance, friction, FSW, forge, explosive, thermit, plasma, laser, EBW, soldering & brazing, HAZ, defects

1. Introduction & Classification of Welding

Welding is a permanent joining process in which two or more parts are joined by coalescence (fusion of the interface) using heat, or pressure, or both, with or without filler metal. Examples: ship hulls, pressure vessels, pipelines, bridges, automobile bodies, rail tracks.

- Advantages: permanent, strong joint (efficiency up to 100 %), any shape/size, low cost, can be automated, portable equipment, can join dissimilar metals, less weight than riveting.
- Limitations: residual stresses & distortion, HAZ with changed properties, skilled labour, fumes & radiation hazards, joint not easily dismantled, defects need inspection.

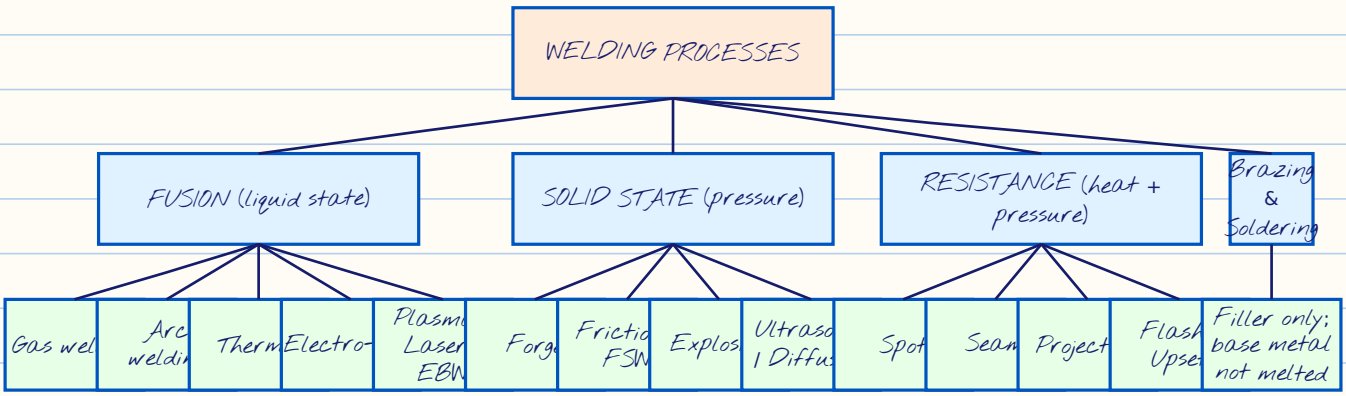


Fig 1.1 Classification of welding processes

Group	Principle	Examples
Fusion welding	Base metal melted (with/without filler), solidifies to form joint, no pressure	Gas, arc, thermit, ESW, plasma, laser, EBW
Pressure / solid-state	Metals heated to plastic state / cold & forced together by pressure; no melting	Forge, friction, explosive, ultrasonic
Resistance welding	Heat by I-squared-R at interface + pressure	Spot, seam, projection, butt (flash)

Brazing / soldering	Filler (m.p. below base metal) melts & flows by capillary action	Torch, furnace, dip, induction

2. Types of Welded Joints & Characteristics

Five basic joint configurations: butt, lap, corner, tee (T) and edge joints. Welds made on these joints are groove, fillet, plug/slot, spot, seam, and surfacing welds.

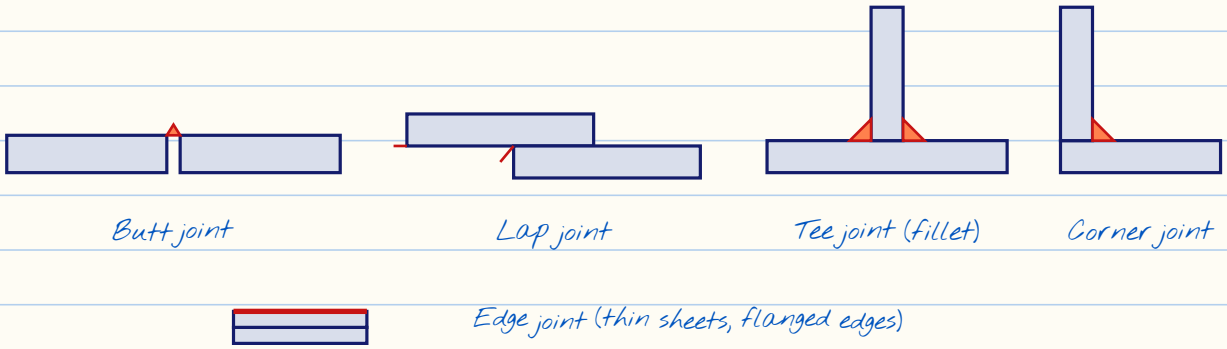


Fig 2.1 Basic weld joints (red = weld metal)

Joint	Characteristics / use
Butt	Two plates in same plane, edges butted. Strongest, best fatigue strength; needs edge preparation for thick plates (square, single/double V, U, J, bevel). Used in pipes, pressure vessels, ship plates.
Lap	Overlapping plates, fillet welds on one/both sides. Simple, no edge prep; strength 70-80 % of butt; eccentric load so bending; used for thin sheets, body panels.
Tee (T)	One plate perpendicular to another; fillet or groove weld; used for stiffeners, frames, beams.
Corner	Two plates meeting at right angle at edges - boxes, frames, tanks.
Edge	Plate edges aligned/parallel, weld along edges - sheet metal, light duty, no load carrying.

- Groove types: square ($t < 5$ mm), single-V (up to 20 mm, 60-90 degrees), double-V (thick plate, less distortion), U / J (very thick - less filler), bevel.
- Fillet weld: triangular section; leg size, throat = 0.707 x leg; strength = 0.707 x leg x length x allowable shear stress.

3. Gas Welding (Oxy-Acetylene)

Heat is produced by burning a fuel gas (acetylene) in oxygen. The flame melts the edges; a filler rod is added. Most common: oxy-acetylene (flame up to ~3200 C). Other fuels: hydrogen, LPG, propane, natural gas (lower temperature - for brazing, soldering).

- Equipment: oxygen & acetylene cylinders (O_2 - black / acetylene - maroon; acetylene dissolved in acetone, max 1.5 bar), pressure regulators, hoses (red-acetylene, black/blue-oxygen), blow-pipe (torch) with mixing chamber & nozzle, flash-back arrestors, goggles, spark lighter.
- Reaction: $C_2H_2 + O_2 \rightarrow 2CO + H_2$ (primary, inner cone, 3200 C); then $2CO + H_2 + 1.5\ O_2 \rightarrow 2CO_2 + H_2O$ (secondary, outer envelope).
- Systems: high-pressure (> 0.1 bar acetylene - cylinder) and low-pressure (injector type, from generator).
- Techniques: leftward (forward) for thin plates (< 5 mm); rightward (backward) for thicker plates (> 5 mm) - better heat use, less filler; vertical & overhead positions.
- Flux: borax for brass/ bronze, C.I., Al flux, to remove oxides.

Types of oxy-acetylene flames

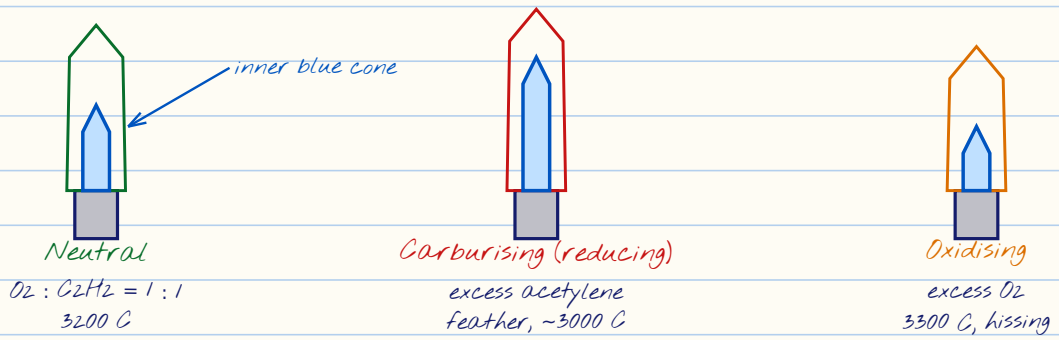


Fig 3.1 Oxy-acetylene flames

Flame	Appearance	Uses
Neutral (1:1)	Clear sharp inner cone, no feather; soft	Most welding: mild steel, C.I., stainless, Cu, Al - no effect on molten metal
Carburising / reducing (excess acetylene)	Three zones; whitish feather around inner cone	Hard surfacing (stellite), monel, nickel, high-carbon steel, Al; low-temp brazing; adds carbon - not for mild steel

Oxidising (excess O_2)	Short, bluish, small cone; hissing sound	Brass, bronze, brazing, zinc-base alloys (forms protective oxide layer); harmful to steel (burning, porosity)

Oxy-acetylene gas cutting

Cutting of ferrous metals by rapid oxidation (burning). Steel is pre-heated to about 800-900 C (Kindlingignition temp.) by neutral flames; a jet of pure oxygen is then directed on the spot: Fe + O_2 -> iron oxide (Fe_3O_4 + heat), which is blown away as slag, forming a kerf. Reactions are exothermic so cutting continues.

- Torch has pre-heating holes around a central oxygen cutting jet. Thickness: up to 300 mm+ steel.
- Works for steels and low-alloy steel; not for C.I., stainless steel, Al, Cu (oxide m.p. higher than metal, or conduction too high) - use plasma cutting / flux injection.
- Conditions: ignition temp < m.p. of metal; oxide m.p. < metal m.p.; low thermal conductivity.

4. Arc Welding - Basic Principles

An electric arc (sustained discharge through ionised gas between electrode & work) provides heat of about 6000-7000 C at the arc column; metal melts at 1500+ C. Arc voltage 20-40 V, current 50-1000 A. Electrode may be consumable (coated rod, wire) or non-consumable (tungsten, carbon).

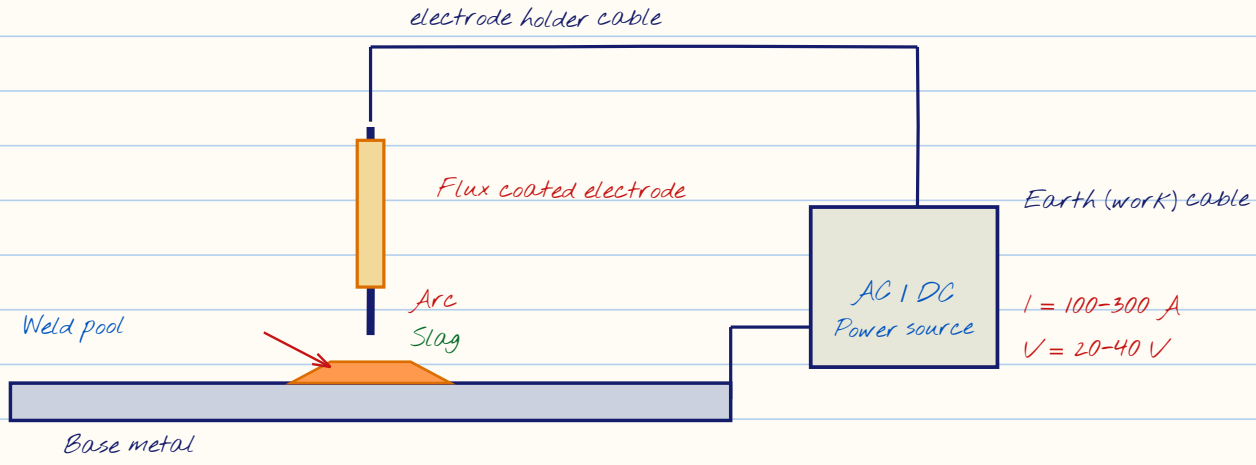


Fig 4.1 Arc welding circuit (SMAW)

- Polarity: DCSP (straight, electrode -ve): 2/3 heat at work - deep penetration, thick plates; DCRP (reverse, electrode +ve): 2/3 heat at electrode - shallow

penetration, thin sheets, Al/Mg (cathodic cleaning of oxide); AC - 50:50, avoids arc blow, for Al with HF.

- Arc blow: magnetic deflection of arc (DC) - reduce by AC, earth connection placement, short arc.
- Heat input: H = (V x I x 60 / S) x eta J/mm (S = travel speed mm/min, eta = 0.7-0.9).
- Power sources: welding transformer (AC, cheap, 20 KVA); rectifier (AC to DC); motor-generator set; inverter (compact, efficient). Duty cycle = (arc-on time / 10 min) x 100.

Power characteristics (V-I curves)

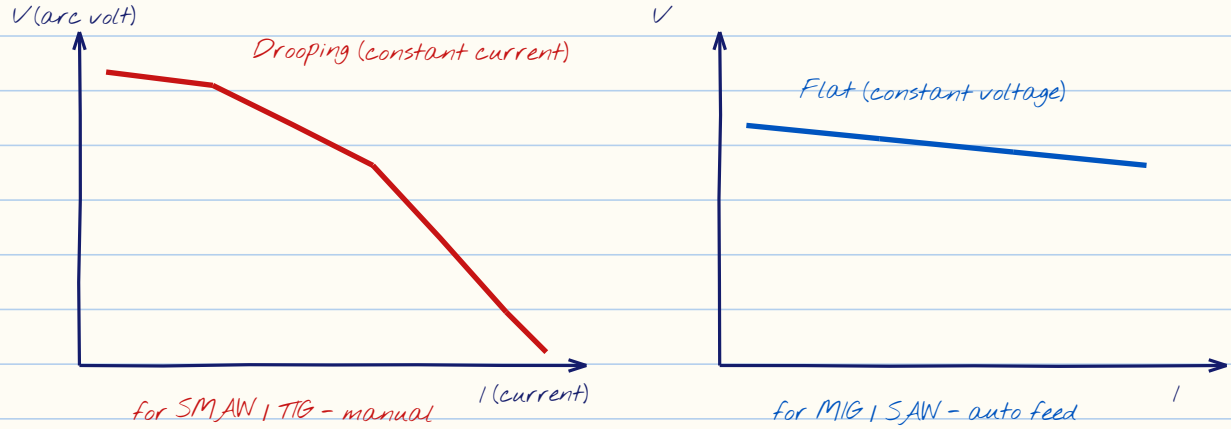


Fig 4.2 Static V-I characteristics

- Drooping (constant-current): large change in arc length (voltage) causes small change in current - manual welder can vary arc length without affecting weld; high OCV (60-80 V) for striking.
- Flat (constant-voltage): arc length self-regulates; current rises sharply if wire stick-out reduces; burn-off adjusts wire feed - used for semi-automatic MIG/CO2/SAW with constant wire feed.

5. Arc Welding Processes

5.1 Manual Metal Arc Welding (MMAW / SMAW / stick)

A flux-coated consumable electrode is held by hand; arc between electrode & work melts both; coating burns to give shielding gas & slag. Electrode specification e.g. E 6013: E = electrode; 60 = tensile strength x 1000 psi (60 Ksi); 1 = position (all); 3 = coating & current type.

Functions of electrode coating (flux)

- Produces shielding gas (CO_2, CO) - protects molten pool from O_2/N_2. - Forms slag that covers weld - slow cooling, protects from oxidation.
- Stabilises the arc (K, Na compounds); adds alloying elements (Mn, Cr) & deoxidisers (Fe-Mn, Fe-Si).
- Improves deposition rate (iron powder), controls bead shape. Types: cellulosic, rutile, basic (low-hydrogen), acid.

- Merits: simple, cheap, portable, all positions, wide range of metals. Demerits: low productivity, stub-end loss (about 50 mm), slag removal, needs skill, fumes.

5.2 Submerged Arc Welding (SAW)

A bare consumable wire is fed continuously into a blanket of granular flux deposited ahead of the arc; the arc burns submerged (invisible, no spatter). Unused flux is recovered by a vacuum; slag peels off.

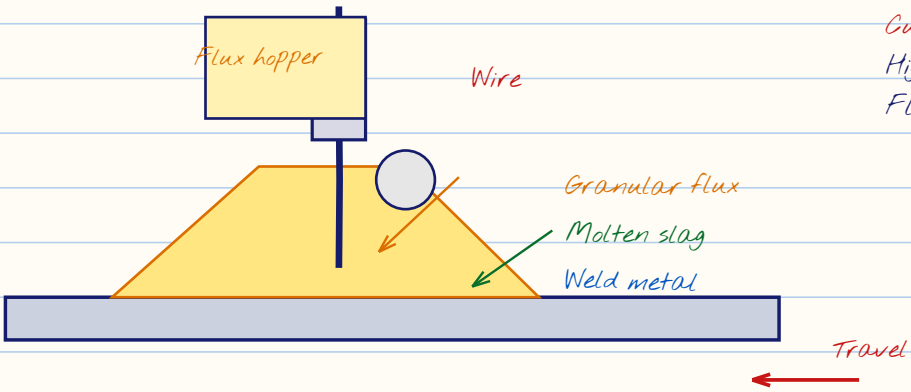
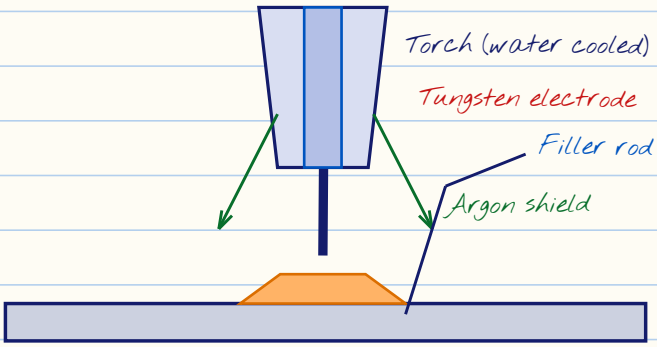


Fig 5.1 Submerged arc welding

- Merits: very high deposition (up to 5x SMAW), deep penetration, smooth uniform bead, no arc flash/spatter, high quality, automatic. Limits: only flat/horizontal, long straight seams; flux handling; thick plate. Uses: pressure vessels, ship panels, pipes, structural girders, tanks.

5.3 TIG welding (GTAW)

Tungsten Inert Gas: arc between a non-consumable tungsten electrode (2 % thoriated / ceriated) and work, shielded by inert gas (Ar / He). Filler rod added separately by hand (optional). DCSP for steel; AC + high-frequency for Al & Mg. Water/gas-cooled torch.

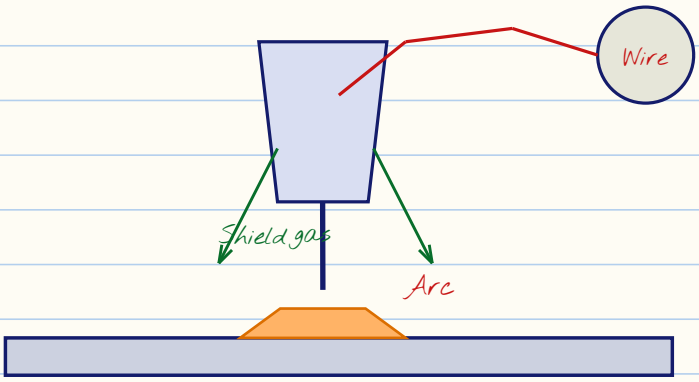


Clean welds, no slag / spatter
Thin sections, Al, Mg, Ti, stainless
Slow, costly, tungsten inclusion risk

Fig 5.2 TIG welding

5.4 MIG welding (GMAW)

Metal Inert Gas (MIG - Ar/He) or MAG (CO_2 / Ar-CO_2 mixes). A continuously fed consumable bare wire electrode with shielding gas; constant-voltage DCRP power. Semi-automatic gun; metal transfer modes: short-circuit (dip), globular, spray, pulsed.



Fast, high deposition, easy to automate
Steel, Al, Cu, stainless; all positions
Gas shield spoiled by wind; costly set-up

Fig 5.3 MIG / MAG welding

Feature	TIG	MIG
Electrode	Non-consumable tungsten	Consumable wire
Filler	Separate rod	Wire itself
Shield gas	Ar, He (inert)	Ar, He, CO_2, mixes
Quality / speed	Very high quality; slow	Good quality; fast
Use	Thin sheet, reactive & exotic metals	Production welding, structures, automobile

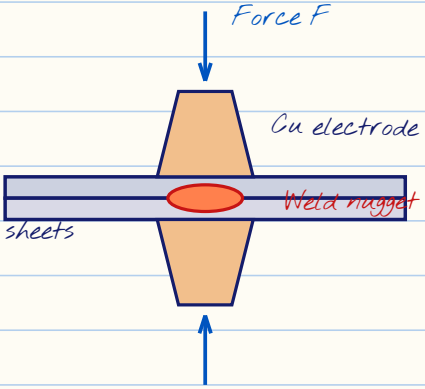
5.5 Electro-slag welding (ESW)

For thick plates (20 mm - 1 m) in vertical position, single pass. Plates are kept with a gap of 25-35 mm; water-cooled copper shoes (dams) both sides. Arc is struck initially under flux; once a slag pool forms, arc extinguishes and heat is produced by resistance heating of conductive molten slag ($I \times I \times R$). Consumable wire (or plate guide) is fed into the pool; weld metal solidifies upwards.

- Merits: single pass for very thick plates, no edge preparation, high deposition (15-20 Kg/h), low distortion. Limits: only vertical, coarse grain & low toughness in HAZ (normalising needed), cannot stop midway. Uses: heavy pressure vessels, ship hulls, heavy machinery frames, bridges.

6. Resistance Welding

Heat is produced by resistance to flow of current across the joint interface: $H = I^2 R t$ (joules). Pressure is applied by electrodes (Cu alloy). Low voltage (1-10 V), very high current (3000-100000 A), time 0.02-few s. No filler, no flux.



Spot welding of sheets
Squeeze - weld - hold - off
Seam: rollers instead of electrodes
(continuous overlapping spots, leak-tight)

Fig 6.1 Resistance spot welding

Type	Description	Use
Spot	Overlapping sheets clamped between two Cu electrodes; nugget forms at interface (3-5 sqrt t mm dia.)	Car bodies, sheet metal, appliances
Seam	Roller electrodes -> continuous seam of overlapping spots	Fuel tanks, drums, gas-tight vessels
Projection	Projections embossed on one part concentrate current; many spots at once	Nuts, bolts, studs on plates
Butt (upset)	Ends pressed together, current heats, upset forgings	Wire, rods, rings
Flash	Arcing (flashing) between ends then forging pressure; metal expelled	Rails, chains, tubes, large sections

- Merits: fast, automatic, no filler, low distortion, clean. Limits: high equipment cost, thickness limit, lap joints only (spot), skilled set-up.

7. Solid-State & Special Welding Processes

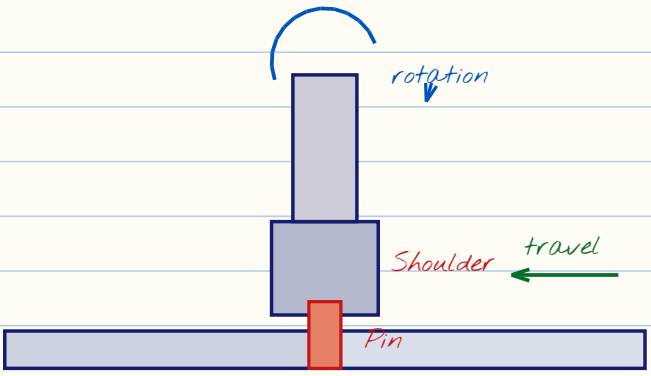
7.1 Friction welding

One part is rotated at high speed against a stationary part under axial force. Frictional heat raises interface to plastic state; rotation stopped and upset force applied -> forged joint, flash expelled. No filler, flux; no melting.

- Types: continuous-drive, inertia (flywheel), linear friction. Rotary - round bars/tubes.
- Merits: strong joint, dissimilar metals (steel-Al, Cu-steel), fast (seconds), no shielding, narrow HAZ. Uses: engine valves, drill bits, shafts, axles, pump parts. Limit: mainly round sections; costly machine.

7.2 Friction stir welding (FSW)

A rotating, non-consumable tool with shoulder and pin is plunged into the joint line of two butted plates and traversed. Friction & plastic deformation heat the metal (below melting); the pin stirs softened material from the advancing to retreating side forming a solid-state joint.



Al, Mg, Cu alloys; aerospace panels
No melting - no porosity, low distortion
Keyhole at end; heavy clamping needed

Fig 7.1 Friction stir welding

7.3 Forge welding

Oldest process: parts heated in a forge to white heat (about 1300 C), overlapped, and hammered / pressed to join (flux - sand/ borax to dissolve oxides). Used for wrought iron, low carbon steel; chains, tools; now replaced by modern methods.

7.4 Explosive welding

A controlled explosive charge drives a flyer plate at high velocity (~ 300-800 m/s) onto a base plate at a small angle (standoff); the oblique impact produces a metal jet that removes oxide films and creates a metallurgical (often wavy) bond - solid state, no melting.

- Joins dissimilar metals not weldable otherwise (Al-steel, Ti-steel, Cu-steel); cladding of large plates, transition joints; tubes to tube sheets. Limit: safety, noise, simple geometry.

7.5 Thermit welding

Exothermic reaction of aluminium powder + iron oxide (thermit): $8Al + 3Fe_3O_4 \rightarrow 9Fe + 4Al_2O_3$ + heat (2500-3000 C). Superheated molten iron is poured into a mould built around the joint; it fuses with the preheated ends. Slag (Al_2O_3) floats. Uses: rail joints, heavy section repairs (shafts, frames), crankshaft.

7.6 Plasma arc welding (PAW)

Constricted arc - gas (Ar) is forced through a small water-cooled copper nozzle producing a high energy-density plasma jet (up to 20000 C). Modes: transferred (arc to work) and non-transferred (arc inside torch). Keyhole mode for thick plates (up to 10 mm single pass).

- Merits: stable arc, narrow weld, deep penetration, high speed, less HAZ; also used for plasma cutting & spraying. Limits: expensive, noise, UV, torch life.

7.7 Laser beam welding (LBW)

A highly focused coherent beam (CO_2 or Nd:YAG / fibre laser, 1-100 KW) melts metal; Keyhole effect gives deep, narrow welds. No vacuum needed; shielding gas used. Heat input low -> small HAZ.

- Merits: precise, non-contact, high speed, automation (robots), welds thin / dissimilar metals, micro-welding. Limits: very high equipment cost, reflective metals (Al, Cu) difficult, joint fit-up must be exact, low efficiency. Uses: automotive tailor-welded blanks, electronics, medical, turbine parts.

7.8 Electron beam welding (EBW)

A beam of high-velocity electrons (30-150 KV) in a vacuum chamber (10-4 to 10-6 torr) strikes the joint; Kinetic energy converts to heat. Keyhole forms - very deep,

narrow weld (depth : width up to 30:1).

- Merits: highest quality, deep penetration (up to 300 mm), minimal distortion & HAZ, reactive metals (Ti, Zr, refractory), no flux/filler. Limits: vacuum chamber (size limit, time), costly, X-ray hazard. Uses: aerospace (jet engine), nuclear, gears, precision parts.

8. Soldering & Brazing

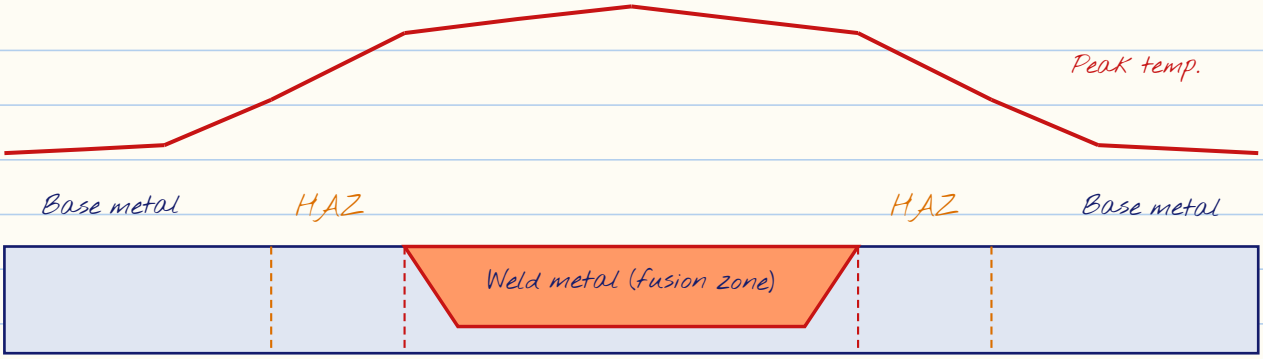
Feature	Soldering	Brazing
Filler temperature	Below 450 C (typically 180-300 C)	Above 450 C, below base-metal m.p. (600-1100 C)
Filler metal	Sn-Pb, Sn-Ag, Sn-Cu (lead-free) alloys	Brass / bronze (Cu-Zn), silver alloys, Al-Si, Cu-P
Flux	Rosin, zinc chloride, ammonium chloride	Borax, boric acid, fluorides
Joint strength	Low (30-50 MPa); electrical contact	Moderate-high (200-400 MPa); load carrying
Joint design	Lap, close fit	Lap/butt; clearance 0.025-0.25 mm for capillary action
Uses	Electronic PCBs, tin cans, radiators, pipes	Tools tips, bicycle frames, heat exchangers, pipes, carbide tips

- Heating methods: torch, furnace, dip, resistance, induction, infra-red. Base metal does NOT melt; filler spreads by capillary action; fluxes remove oxide; clean surfaces essential.
- Braze welding: filler is deposited like a fusion weld (no capillary) - repair of C.I.

9. Heat Affected Zone, Pre/Post Heating

The Heat Affected Zone (HAZ) is the portion of base metal which is not melted but whose microstructure & properties are altered by welding heat.

- Width of HAZ increases with heat input, plate thickness, preheat; decreases with high travel speed & low heat input (laser/EBW have narrowest HAZ).
- Problems: hardening (martensite in steels) -> cracks; grain coarsening -> poor toughness; softening in cold-worked or age-hardened alloys.



HAZ zones: coarse grain (near fusion line) -> fine grain (normalised) -> partially transformed -> unaffected

Fig 9.1 Weld zones and peak temperature

Pre-heating & Post-heating

- Pre-heating (150–300 C+): reduces cooling rate -> less martensite & residual stress, removes moisture, avoids cold (hydrogen) cracking; needed for high-carbon, alloy steels, thick sections, cast iron.
- Post-heating / PWHT: stress relieving (550–650 C for steel), tempering, normalising; reduces residual stresses, improves toughness, hardness & dimensional stability; hydrogen bake-out.
- Carbon equivalent $CE = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15$: higher CE (> 0.4) => weldability poor, preheat needed.

10. Welding Defects - Causes & Remedies

Defect	Causes	Remedies
Porosity / blow holes	Gas trapped (moisture, dirt, oil, wrong electrode, damp flux, long arc)	Clean/ dry, low-H electrodes, shorter arc, correct gas flow, slow cooling
Slag inclusion	Slag trapped between passes; poor cleaning; low current	Clean slag after each pass, proper manipulation, higher current
Lack of fusion	Low current/heat, high speed, wrong angle	Increase current, reduce speed, correct angle & joint design
Incomplete penetration	Narrow root gap, low current, thick root face	Wider gap, higher current, better joint preparation
Undercut	High current, long arc, fast travel	Reduce current/ arc length, steady travel, correct angle
Overlap	Low travel speed, too low current, wrong electrode angle	Increase speed & current

Cracks (hot / cold)	High residual stress, high CE, hydrogen, rapid cooling, S & P	Pre-heat, low-hydrogen electrodes, proper sequence, PWHT, ductile filler
Spatter	High current, long arc, arc blow, damp electrode	Reduce current/ arc, dry electrodes, anti-spatter spray
Distortion / warpage	Uneven heating & contraction	Jigs, tack welds, back-step, symmetrical welding, pre-set, stress relief
Burn-through	Excess heat, thin metal, big root gap	Lower current, backing strip, quicker travel
Crater / poor bead	Sudden arc break; wrong technique	Fill crater, run-off tabs

Important exam questions - Unit II

- Classify welding processes. Types of weld joints & their characteristics. (10 M)
- Oxy-acetylene welding: flames & their uses; gas cutting principle (10 M).
- Arc welding: principle, V-I characteristics, polarity; MMAW electrode coating (10 M).
- Explain SAW / TIG / MIG / ESW with neat sketch & applications (5-10 M each).
- Resistance welding types; friction, FSW, explosive, thermit, plasma, laser & EBW (5 M each).
- Brazing vs soldering; HAZ; pre & post heating; welding defects, causes & remedies (10 M).